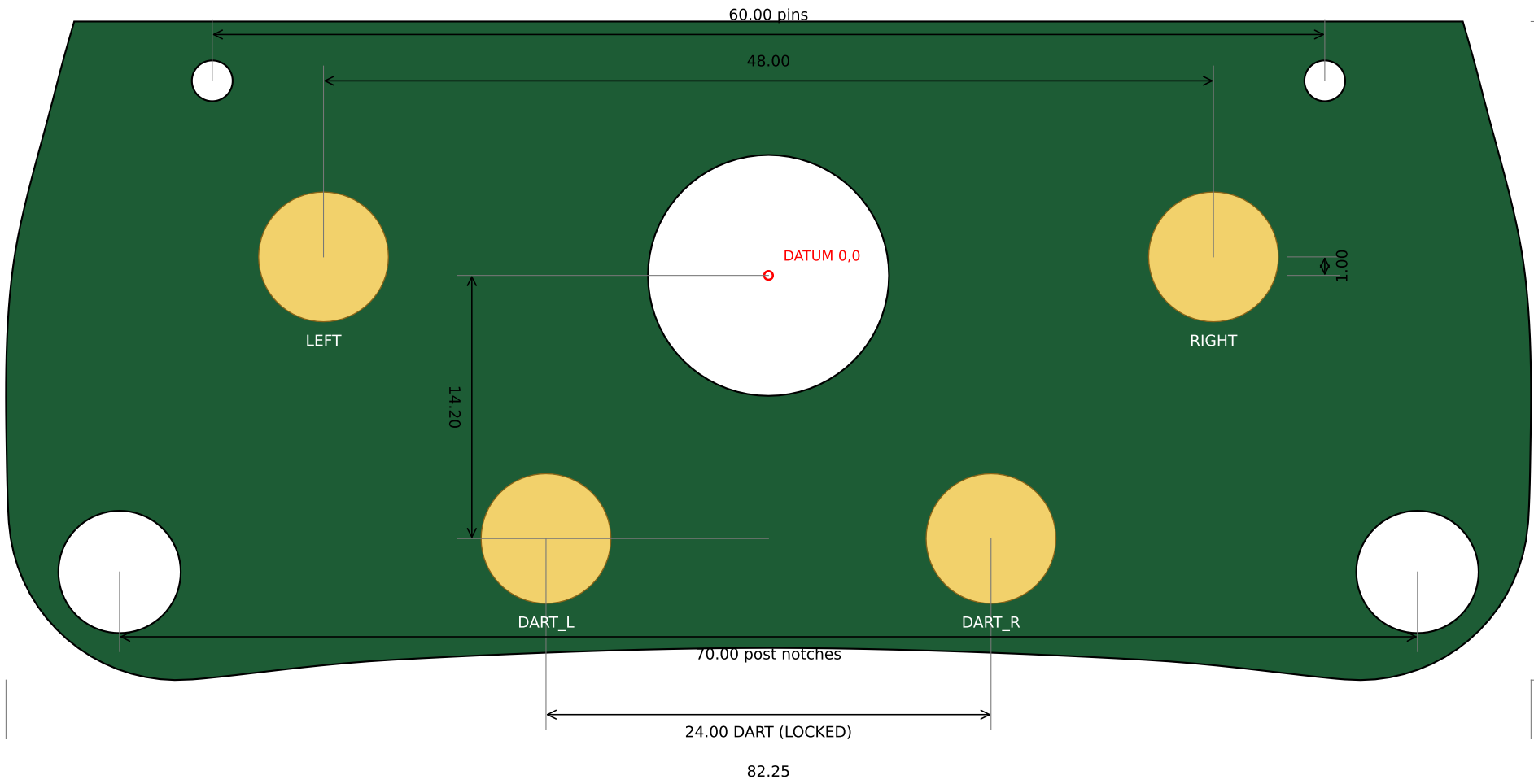


STRUTHIO-CP1 control PCB

two-layer contact board under STRUTHIO-CM1: four interdigitated pads, GND bus, 5-wire pad row

FRONT VIEW (contact side, faces the mat)



STRUTHIO-CP1 82.2 x 35.5 mm, 1.0 mm FR-4, ENIG

BOARD

FR-4, 2 layers, 1.0 mm, 1 oz copper

Finish ENIG (contacts must be gold, no HASL)

Solder mask both sides; white legend on the back only

Size 82.25 x 35.53; outline, window and notches in Edge_Cuts

CONTACTS (front)

4 interdigitated pads dia 7.0, fingers 0.30 / gaps 0.30

Centres = CM1 pill centres; DART pads 24.00 apart (LOCKED)

Mask opening = pad + 0.10; keep contacts clean (no flux, no silk)

NETS -> ESP32-S3 (Waveshare 3.5B)

LEFT -> GPIO17 (LEFT WING)

RIGHT -> GPIO18 (RIGHT WING)

DART_L -> GPIO21 (DART LEFT)

DART_R -> GPIO38 (DART RIGHT)

GND -> board GND (common)

All active low; internal pull-ups in firmware v0.12

WIRE PADS (back)

5 x plated dia 1.0 / pad 1.8, 2.54 pitch, left to right (front view):

GND, LEFT, DART_L, DART_R, RIGHT

~24 AWG; W1 white L, W2 blue R, W3 yellow DL, W4 orange DR, W5 black GND

MECHANICAL

2 x NPTH dia 2.2: carrier heat-stake pins

Notches R3.3 clear the lower case-screw posts

Window dia 13.0: speaker path (CP1 is part of the baffle; seal ring on the back)

PUI speaker gasket seats on the back face around the window

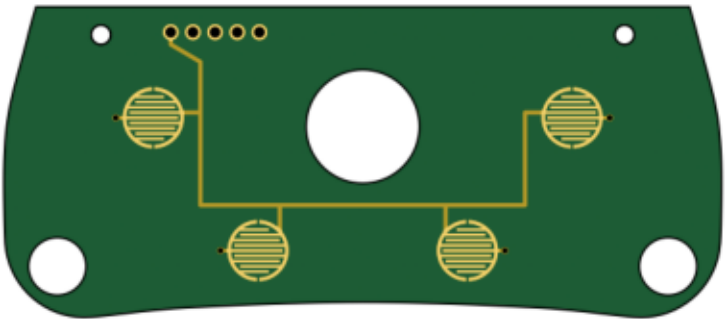
FILES

cad/handheld/cp1/cp1_gerbbers.zip (Gerber RS-274X + Excellon, DRC checked)

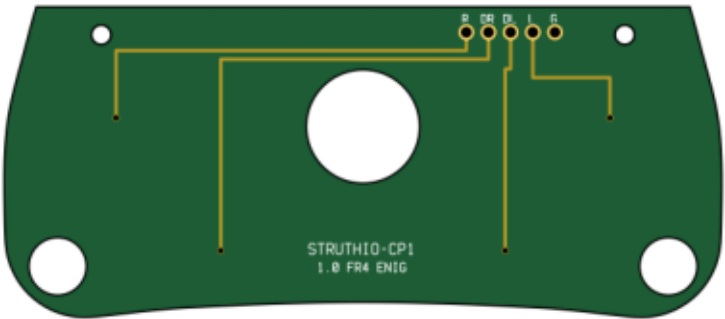
cad/handheld/dxf/cp1_outline.dxf

Regenerate: python3 cad/handheld/cp1/make_cp1.py

FRONT (contacts, faces the CM1 mat)



BACK (seen from behind)



PART	STRUTHIO-CP1 rev A
PROJECT	STRUTHIO handheld
SOURCE	cad/handheld/struthio_handheld.scad, checked by check_handheld.py
UNITS	mm; datum = speaker-window centre; general tol. +/-0.15
OWNER	R.A. Peddycoart